Artifact: A Reproducible, Database-Free Pipeline for Detecting Research-Software Paradigms in the DELFI Community

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Abstract: This artifact accompanies the DELFI 2026 paper *Research Software Engineering in the Learning Technologies: a replication study of research paradigms in the DELFI community*. It provides a self-contained, file-based pipeline that reproduces the study's label aggregation, inter-coder reliability (ICR) metrics, descriptive figures and secondary bias analysis directly from the shipped intermediate data — *without* access to the copyrighted DELFI paper PDFs or to a commercial LLM API. The six pipeline steps are orchestrated by a single management script; the two steps that require the paper texts (preprocessing, human re-annotation) and the one that requires an LLM API token (the experiments) are skipped by default. A programmatic guard enforces that no paper full text is ever contained in the released data. We describe the artifact's structure, how to run it, the reproducibility guarantees, and the constraints that motivated its design for the artifact-evaluation track.

Resource Type: Software (with an accompanying data set)
License: Code: MIT; Data: CC-BY-4.0
Stable ID: *archival DOI to be minted on Zenodo before camera-ready* (e.g. `10.5281/zenodo.XXXXXXX`)
Repository: https://github.com/juliandehne/rse_in_delfi_artifact

Keywords: Research Software Engineering, replication, large language models, inter-coder reliability, reproducibility, DELFI

1 Introduction

The companion study [DLW26a] replicates and extends a 2017 analysis of research-software paradigms across the DELFI proceedings, using three open-weight large language models (LLMs) to annotate $n = 798$ papers and validating the result against the earlier human-coded reference study. This artifact packages the *evidence and the computation* behind those results so that a reviewer can re-run the analysis end-to-end on a laptop in minutes.

The central design constraint is legal: the analyzed DELFI papers are copyrighted and may not be redistributed. The artifact therefore ships only derived data (LLM labels, justifications, human re-annotations and bibliographic metadata) and is explicitly built so that the two paper-dependent steps and the API-dependent step can be skipped while still reproducing every published number and figure.

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2 Artifact overview

The artifact is a Python package (`pipeline/`: `config`, the six steps and a full-text guard), the management script `run_pipeline.py`, the shipped intermediate data (`data/`) and this paper (`artifact-paper/`). It is database-free: whereas the original study stored papers in MySQL, every step reads and writes CSV files under `data/`. We target the *Available* and *Reusable* badges: the package is archived, documented and parameterised so it can be re-run on new inputs.

3 The pipeline

The pipeline has six steps (Table 1). Steps 3, 5 and 6 run from the shipped data and constitute the default reviewer run; steps 1, 2 and 4 are *skipped by default* because they need resources reviewers do not have.

Tab. 1: Pipeline steps and default execution policy.

#	Step	Default	Requires
1	Preprocessing (PDF → text)	**skipped**	paper PDF folder
2	LLM experiments	**skipped**	SAIA/KISSKI API token
3	Descriptive analysis (figures)	runs	shipped data
4	Human re-annotation	**skipped**	paper PDF folder (interactive)
5	Label aggregation (intra + inter, ICR)	runs	shipped data
6	Secondary analysis (paradigm biases)	runs	shipped data

Data flow. Step 1 extracts text from the PDFs; step 2 annotates each paper with three LLMs (three runs each) via the SAIA API. Step 5a aggregates the runs per model (majority vote) with intra-model ICR; step 4 elicits a human label where all three models disagree; step 5b aggregates across models (majority vote, ties broken by the human label), reports cross-model ICR and validates against the 2017 reference study. Steps 3 and 6 produce the figures and the bias analysis. `run_pipeline.py` runs the default steps; `--paper-folder` enables steps 1/4 and `--run-experiments` (with a SAIA token) enables step 2. The no-full-text guard runs before and after every invocation.

4 Reproducing the results

```
python -m pip install -r requirements.txt
python run_pipeline.py            # steps 3, 5, 6 from shipped data
```

This regenerates `data/inter_LLM/df_llm_experiments_final_aggregated_*.csv` (798×185), the ICR tables, the paradigm-distribution and label-trend figures, and the combined-label bias tables.

5 Reproduced results

Step 5, run on the shipped data, recomputes the cross-model inter-coder reliability as Krippendorff's α [Kr04] (binary labels complemented by Gwet's AC1 [Gw08] in the released tables): average α = 0.51 for the binary labels, 0.66 for the ordinal labels and 0.60 overall, with α = 0.33 against the 2017 reference study (binary). All three models disagreed completely on 0, 6, 17 and 9 papers for the process, outcome, design and acceptance paradigms respectively; those cases were resolved with the shipped human labels.

6 Data and the no-full-text guarantee

The released data contain only labels, LLM/human justifications and bibliographic metadata. The original annotation script dropped the `abstract`, `text` and `references` columns before persisting results "for legal reasons"; the artifact preserves and enforces this invariant programmatically: `pipeline/fulltext_guard.py` scans every shipped table and fails if a banned full-text column is present or any cell exceeds a length threshold incompatible with a short justification. Step 1's full-text output goes only to a `.gitignore`'d directory.

7 LLM documentation

Per the DELFI AET LLM policy, Table 2 lists the three open-weight models used in step 2, all served through the KISSKI SAIA API. Each model was run three times (run_1–3). All runs used identical, deterministic inference parameters: **temperature** = 0, **top_p** = 1.0, **seed** = 42, no **stop sequences**, no explicit **max-tokens** cap (provider default), a 300 s request timeout, and no function/tool calling (plain JSON-mode prompting). The models are open-weight, so a closed-weight snapshot date does not apply; the snapshot is pinned by the version in the model identifier, and the run dates are recorded in the result filenames.

Tab. 2: LLM models and versions used for annotation (KISSKI SAIA API).

Short name	Model identifier (version)	Runs
mistral	`mistral-large-3-675b-instruct-2512`	3
llama	`llama-3.1-sauerkrautlm-70b-instruct`	3
gemma	`gemma-3-27b-it`	3

Prompts. The complete prompt — the system prompt and the user-prompt template with its `{row[...]}` placeholders — is shipped verbatim at `data/prompt_templates/prompt_template_1.md`. The placeholders `title`, `authors` and `year` resolve to public bibliographic metadata; the placeholders `abstract`, `text` and `references` resolve to the copyrighted paper content and are therefore the *only* part of a fully resolved prompt that cannot be redistributed. The substitution logic (`pipeline/step2_experiments.py`, `fill_user_prompt`) is included so the resolved prompt can be regenerated by anyone holding the paper PDFs.

8 FAIR preparation and targeted badges

The artifact targets the **Available** and **Reusable** badges: *Findable/Available* via a stable archival DOI (Zenodo, from a tagged release) and machine-readable metadata (`CITATION.cff`); *Accessible* via open licensing (MIT code, CC-BY-4.0 data; `LICENSE`, `LICENSE-data`); *Interoperable* via open CSV with a documented data dictionary (`data/README.md`); and *Reusable* via a parameterised, documented, re-runnable pipeline. No personal or human-subject data are released, so no further DSGVO/ethics statement beyond the no-full-text guarantee is required.

9 Requirements and environment

Python 3.11+, `pandas`, `numpy`, `scikit-learn`, `krippendorff`, `statsmodels`, `matplotlib` and `openpyxl`; see `requirements.txt`. `pymupdf` is needed only to re-run step 1, and an OpenAI-compatible client plus a SAIA token only to re-run step 2.

10 Limitations

Steps 1 and 2 cannot be exercised by reviewers without the copyrighted PDFs and an API token; their outputs are shipped and verified downstream instead. LLM annotations are sampled at temperature 0 with a fixed seed, but exact reproducibility of step 2 ultimately depends on the provider's model snapshots.

References

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